Future Road Map of Software Development of PWD

PPMS

Sl	Module	Present Status	Future Modification Plan
1	Upcoming Projects	At present, the system accommodates only ongoing and newly approved projects and does not provide any functionality to register, track, or manage upcoming projects prior to approval.	To address this limitation, the following features may be introduced: 1. Listing and Tracking of Upcoming Projects before formal approval. 2. End-to-end planning support from the inception of the Development Project Proposal (DPP) through its approval. 3. Centralized record-keeping of all documents generated during the planning and approval process for future reference. 4. Early linkage of all concerned offices related to the project to facilitate coordination and preparedness.
2	Preliminary Procurement Plan	After procurement packages are attached by the Project Division, the concerned field offices will notify the system of any implementation constraints. If no constraints exist, the field offices shall prepare the Preliminary Procurement Plan (PPP) by indicating estimated timelines for key stages, including soil investigation, design submission, and contractor selection.	1. Upon the attachment of procurement packages, Project Division will prepare Phase-1 of the Preliminary Procurement Plan (PPP) by defining design-type-based timelines, from report/design submission up to estimate preparation. 2. The concerned field office will be linked and automatically notified. 3. The field office then reviews the package, reports any implementation constraints, and—if none exist—finalizes the remaining part of PPP.
3	Dashboard	At present, all users view a generic/common dashboard on their home page that lists all upcoming tasks. Due to the lack of task prioritization, users often face difficulty in identifying immediate or critical activities among multiple pending tasks. Furthermore, the system does not include a dedicated task management mechanism, preventing Heads of Offices from formally assigning tasks to subordinate officers through the system. The absence of an in-system commenting or chat feature among project-related officers also makes inter-office coordination difficult and results in a lack of documented communication records.	1. Under the proposed system, users will view a priority-driven dashboard displaying immediate pending tasks, with deadline-critical activities clearly marked. 2. A structured task management system will be introduced, enabling Heads of Offices to assign, track, and monitor tasks for subordinate officers within PPMS. 3. In addition, an integrated commenting and chat system will be implemented to facilitate inter-office communication while ensuring proper audit trails and communication evidence. 4. The system may further support customized dashboards tailored to the specific functional requirements of Field Offices, Supporting Offices, Monitoring Offices, MIS, and the Chief Engineer's Office.

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4	Work Program	Currently, the system includes three predefined work programs consisting of 18 items, categorized based on the number of floors of the structure. Field offices can select any one of these work programs according to the building height and modify it as required. Once the work program is finalized, the system allows field offices to upload site progress information along with photographs.	1. The work program should be automatically generated based on BOQ items (manpower, materials, and tools) derived from the Estimating Software. The generated work program shall define activity-wise timelines based on calculated quantities of manpower, materials, and tools. Field offices will have the flexibility to adjust these timelines based on site realities before finalization. 2. After finalization, BOQ items shall be automatically mapped to corresponding work program stages, enabling field-level progress updates with photographs for each item. This will facilitate realistic physical progress tracking. 3. Additionally system can automatically generate photo-based progress presentations for reporting and review purposes.
5	Package Split and Merge System	At present PPMS has its package manipulation system, but the system needs to be modified.	1. The System is required to be easy to user field office. 2. The approval process of the split-merge requests should be introduced in the system.
6	Progress Report	Already Done but yet to be integrated with the system.	

Estimating Software

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1	Repair Estimate	Field Offices can upload the approved Annual Procurement Plan (APP) into the system and generate repair estimates against the approved works included in the APP. The system also supports forwarding repair estimates to higher offices for review and approval.	1. Enable end-to-end digital approval of APPs, from Field Division up to the Coordinating Office. 2. Introduce functionality for preparation and approval of Revised Annual Procurement Plans (RAPPs) within the system. 3. Provide mechanisms to record tender-related information and store photographic evidence of damaged and repaired items for documentation and monitoring. 4. Integrate the system with the BIMH Database to accurately identify and link structures undergoing repair works.

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2	Measurement From Drawing	At present, no dedicated Measurement from Drawing module exists within the system.	1. The system shall enable automatic sharing of drawings uploaded in PPMS with the Estimating Software, or alternatively allow direct uploading of drawings within the Estimating Software, accessible through PPMS. 2. Support multi-format drawing input, including image, PDF, and DWG files. 3. Provide tools to take measurements directly from drawings and seamlessly transfer the measured data to the estimating module. The system shall also allow users to edit or remove measurements, both at the drawing level and within the generated estimate. 4. Ensure bi-directional linkage such that selecting a measurement item in the estimate highlights the corresponding element in the drawing. 5. Preserve all measurement take-off data and associated drawings in the system for audit, review, and future reference.
3	Rough Estimate	This Module is already existing in the DPP Software.	*[Not Clear what are the requirement]*
4	Special Package Modules	At present, only a limited number of package modules are available in the system (i.e. Tubewel). These modules are not mandatory or integral components of the software; rather, they function as optional features intended to improve efficiency and save time.	The following package modules may be considered for inclusion in the system as optional, efficiency-enhancing features: a) Boundary Wall Module (PWD Design Division Provides Designs) b) Sewage Treatment Plant (STP) Module (no approved PWD guideline/design standard currently available) c) Rainwater Harvesting Module (no approved PWD guideline/design standard currently available) d) Articulated LPG System Module (no approved guideline currently available)

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5	SoR	At present, the Estimating Software does not support in-system preparation of the Schedule of Rates (SoR). Instead, SoR data is manually prepared outside the system and subsequently fed into the software by the programmer.	1. A dedicated Schedule of Rates (SoR) module may be developed to enable the concerned office or authority to prepare, manage, and publish SoRs directly within the system. The module shall allow SoR generation from basic item rates, with full flexibility to add new items, modify existing items, or safely deactivate obsolete items. 2. To ensure data integrity, the system shall maintain version control and dependency management, so that any modification to the SoR does not affect previously prepared or approved estimates within the system. 3. Moreover, the system shall facilitate direct inclusion of approved non-scheduled items by authorized users without requiring programmer intervention. 4. The system shall provide a facility to upload, store, and associate reference images with each SoR item for identification and clarity.
6	Special Package Modules	At present, only a limited number of package modules are available in the system (i.e. Tubewel). These modules are not mandatory or integral components of the software; rather, they function as optional features intended to improve efficiency and save time.	The following package modules may be considered for inclusion in the system as optional, efficiency-enhancing features: a) Boundary Wall Module (PWD Design Division Provides Designs) b) Sewage Treatment Plant (STP) Module (no approved PWD guideline/design standard currently available) c) Rainwater Harvesting Module (no approved PWD guideline/design standard currently available) d) Articulated LPG System Module (no approved guideline currently available)
7	Other minor Changes	a) At present, the PE can assign only one SDE to an estimate, even when the estimate includes both Civil and E/M components.	a) Extend the system to allow assignment of multiple SDEs to a single estimate to support multi-disciplinary estimates (Civil, E/M, etc.).
		b) At present, all estimates are prepared centrally, with each supporting office contributing its respective portion of the estimate.	b) Provide functionality for the Project Division to declare an estimate as either centrally prepared or fully prepared by the Field Office.

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		c) System generates pdf and excel/csv file of estimate only. the system cannot generate full estimate documents in a single pdf/doc.	c) Introduce a comprehensive estimate documentation facility that includes automatic memo numbering during forwarding, an Estimate Committee declaration checklist, and the ability to attach relevant drawings and supporting documents, enabling the complete estimate to be generated and downloaded in a single PDF format.
		-	d) OTP verification of the authorized officers during estimate forward and approval. Notifying users of any added estimate via email and SMS.

Bill Variation Module (Proposed by Software Team)

Sl	Item name	Feasibility	Remark
1	Tender From Existing Estimate	This module is not practically feasible unless the e-GP system provides API access to PWD for data exchange.	Currently not implementable
2	BoQ Management	Automatic inclusion of tenderers' rates from e-GP is not possible without direct API access to the e-GP database. Therefore, manual entry of tenderers' rates is required, which has already been addressed within the PPMS scope.	Already been addressed within the PPMS scope.
3	Progress Entry	This functionality has already been covered and proposed under the PPMS Progress Entry module.	Already been addressed within the PPMS scope.
4	MB, RMB and Bill Generation	Measurement Book (MB), Revised Measurement Book (RMB), and Bill generation can be derived from field-level progress data. This functionality is currently unavailable and requires new development.	This Module is need to be developed
5	Managing Variation Item	Variation item management is functionally similar to estimate preparation. Therefore, this module does not require development from scratch and can be implemented by extending existing estimation functionalities.	No need to develop from scratch.
6	Variation Approval System	An estimate approval workflow already exists within the system. As such, a separate variation approval module does not need to be developed from scratch and can be integrated with the existing approval mechanism.	No need to develop from scratch.